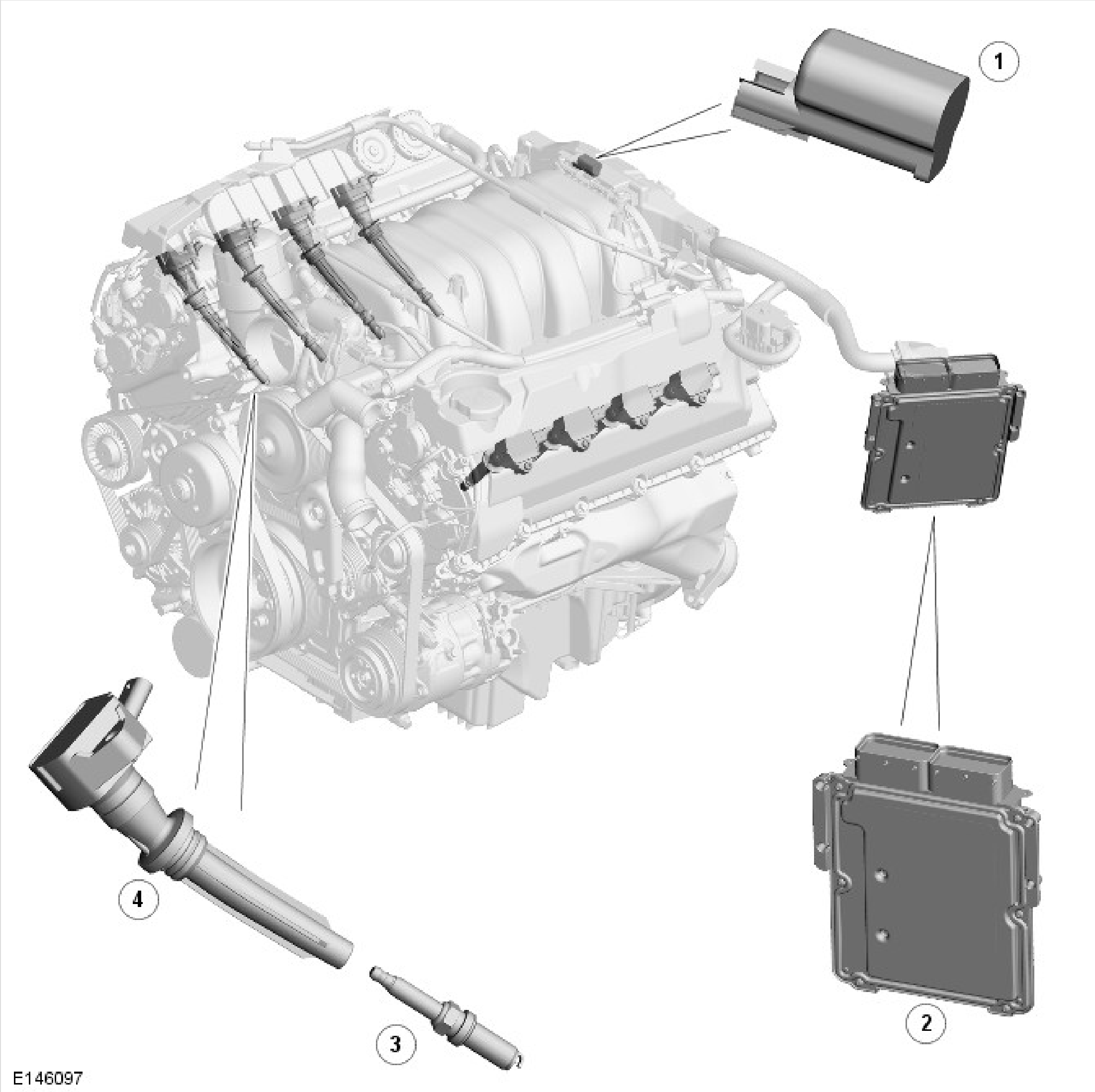
PUBLISHED: 23-AUG-2012
2014.0 RANGE ROVER (LG), 303-07

ENGINE IGNITION - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

COMPONENT LOCATION

NOTE:

Installation on naturally aspirated engine shown, installation on supercharged engine similar.



ITEM	DESCRIPTION
1	Radio frequency interference suppressor
2	Engine control module
3	Spark plug (8 off)
4	Ignition coil (8 off)

OVERVIEW

The engine ignition system is a coil-on-plug, single spark system controlled by the ECM (engine control module) . An iridium tipped spark plug is installed in each combustion chamber, between the inlet and exhaust valves, and an ignition coil is installed on each spark plug. A RFI (radio frequency interference) suppressor is connected to the power feed to the ignition coils.

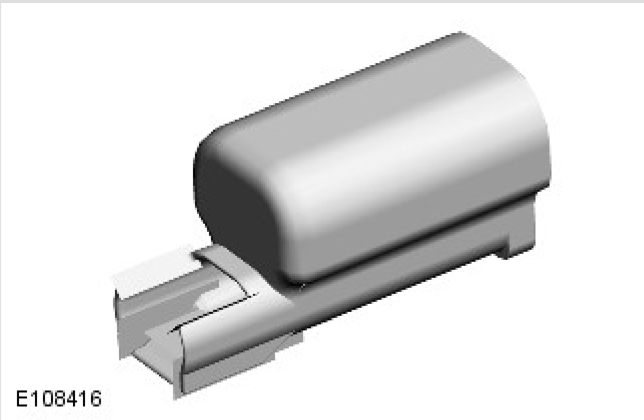
DESCRIPTION

IGNITION COILS

The ignition coils are installed in the cylinder head covers, under the NVH (noise, vibration and harshness) covers. Each ignition coil locates on a spark plug and is secured to the related cylinder head cover with a single screw. Each ignition coil incorporates a three pin electrical connector for connection to the engine harness.

Each ignition coil contains a primary and a secondary winding. The primary winding receives electrical power from the ignition relay in the power distribution box. A power stage in the primary winding allows the ECM to interrupt the power supply, to induce a voltage in the secondary winding and thus the spark plug. A diode in the ground side of the secondary winding reduces any undesirable switch-on voltage, to prevent misfiring into the intake manifold. The power stage limits the maximum voltage and current in the primary winding, to protect the power stage and limit the voltage in the secondary winding.

RFI SUPPRESSOR



The RFI (radio frequency interference) suppressor is installed on the engine harness carrier at the rear of the engine.

OPERATION

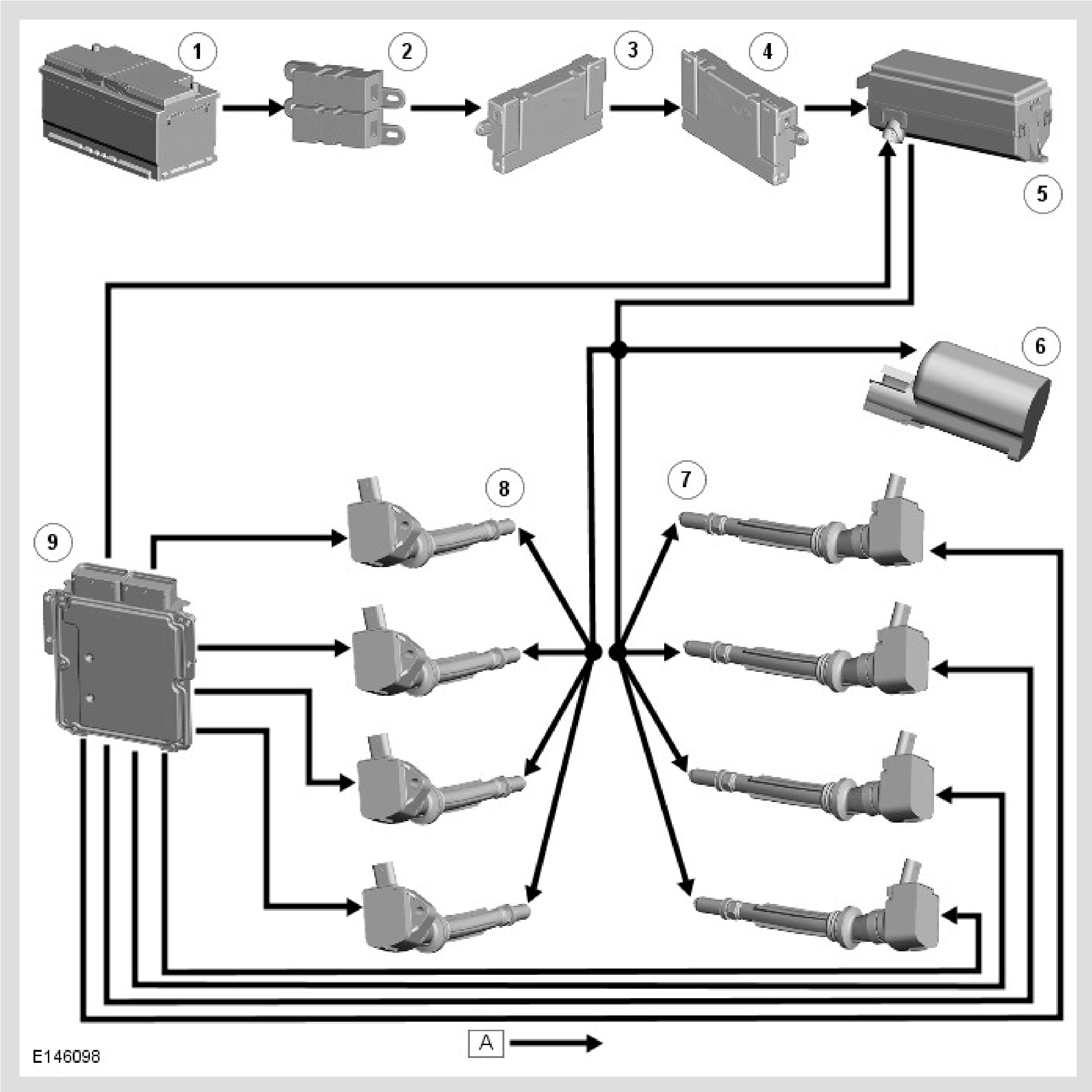
The ignition coils are supplied with electrical power from the ECM relay in the EJB (engine junction box) . The ECM controls the operation of the ECM relay.

The ECM sends a separate signal to each ignition coil to trigger the power stage switching. The ECM calculates the dwell time from the battery voltage and engine speed, to ensure a constant energy level is produced in the secondary coil each time the power stage is switched. This ensures sufficient spark energy is available without excessive primary current flow, which avoids overheating and damage to the ignition coils.

The ECM calculates the ignition timing for individual cylinders from:

- Engine speed
- Camshaft position
- Engine load
- Engine temperature
- The knock control function
- The shift control function
- The idle speed control function.

CONTROL DIAGRAM



A = HARDWIRED.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Auxiliary junction box
5	Engine junction box (ECM relay)
6	RFI suppressor
7	Left cylinder bank ignition coils (4 off)
8	Right cylinder bank ignition coils (4 off)
9	Engine control module